The plant-parts atlas

Teacher guide and worked answers

Identify roots, stem, leaves and flower from plant structures, while distinguishing hidden from absent.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Summer | C30

Identify the parts of a flowering plant (roots, stem, leaves, flower).

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Plants are living things.

A diagram represents selected features, not every detail.

Success evidence

Name the four main parts.

Locate parts on a different plant shape.

Use “not visible” when a view hides evidence.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Optional identified non-toxic potted plant; no uprooting required.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Practical care

Use an adult-selected non-toxic plant if available. Do not taste plant material or touch eyes; wash hands after handling.

Do not uproot, cut or pull apart a live specimen. The supplied diagram gives a complete root view.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Make a guide another pupil can use

Slide	Stage	Minutes	Focus
2	Retrieval	2-5	Notice the structure
3	I do	5-12	I identify the part, then label it
4	I do	Within stage	I keep names when shapes change
5	I do	Within stage	I separate seen from hidden
6	We do	12-16	Atlas partners: point, name, check
7	We do	16-20	The observation editor
8	Hinge	20-23	A plant has small green flowers. What follows?
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Make a guide another pupil can use

Explain that today is about accurate identification; the next lesson explores jobs in detail.

2 Retrieval Notice the structure

1 yes. 2 accept any correct part. 3 yes, opaque containers can hide roots. Avoid awarding a guessed hidden detail as an observation.

3 I do I identify the part, then label it

Think aloud: “I trace A below the soil. I call these roots. I follow B up from them: this is the stem. I find C attached to the stem, then D.” Use the labels and point to leader endpoints.

4 I do I keep names when shapes change

Shares I do time. Think aloud: “Colour alone cannot identify the part: the small green structure can be a flower. I combine position and structure.” This is a simplified comparison, not a botanical species key.

5 I do I separate seen from hidden

Shares I do time. Think aloud: “I report what this view shows. I cannot conclude that a flowering plant will never flower just because it has none today.”

6 We do Atlas partners: point, name, check

W1: A roots, B stem, C leaves, D flower. Reveal labels one at a time in the interactive route. Paper route uses the lettered plant diagram and observation E.

7 We do The observation editor

W2: no roots visible is supported. The opaque pot hides them, so no roots at all is not justified. Ask pupils to improve the flower claim in the same way.

Reveal: Report “roots hidden” and “no flower visible today”.

8 Hinge A plant has small green flowers. What follows?

Correct second option. If pupils choose first, revisit the two distinct green structures.

A. All its green parts must be leaves. - Stems and some flowers can be green too.

B. It can still have roots, stem, leaves and flowers. - Part names do not depend on bright flower colour.

C. Its hidden roots do not exist. - A blocked view does not prove absence.

9 You do Choose your science route

Set ONE route. Assess correct placement or clear verbal location, not drawing skill. Allow a pupil to direct an adult drawing. Keep identification separate from later functional explanations.

10 You do Make your own evidence record

Assess correct placement or clear verbal location, not drawing skill. Allow a pupil to direct an adult drawing. Keep identification separate from later functional explanations.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Assess correct placement or clear verbal location, not drawing skill. Allow a pupil to direct an adult drawing. Keep identification separate from later functional explanations.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Use part names with location clues; say hidden or not visible when evidence is limited.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Use part names with location clues; say hidden or not visible when evidence is limited.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 yes. 2 any correctly named plant part. 3 yes.

We do W1-W2

W1 A roots, B stem, C leaves, D flower. Location clue: roots branch below the soil, leaves attach along the stem, or flower has petals in this example. W2 roots are hidden by the pot and no flower is visible today; do not infer absence or never flowering.

Hinge

Second option correct. Green colour alone is not a part identifier; hidden roots cannot be judged absent.

S1-S5

S1 roots. S2 stem. S3 leaves and flower respectively. S4 roots hidden. S5 any correctly named part with appropriate position if drawn.

M1-M4

M1 roots connected below stem, leaves attached to stem, flower attached above; accept variations. M2 flat structure attached to stem, combined with other visible clues. M3 stem or some flowers. M4 e.g. roots cannot be seen inside the pot; no flower is visible at this time.

T1-T4

T1 stem and leaves. T2 the base is outside the photograph. T3 no open flower is visible in this photograph. T4 e.g. a wider whole-plant view including base and a permitted root diagram or transparent-container view; a close view of an open flower at another appropriate time. Do not require uprooting.

Exit E1-E3

E1 roots. E2 stem. E3 no; flowering depends on species, maturity and season, among other conditions.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Secure core: identify all four parts on the atlas and distinguish a hidden part from an absent part. If names are secure but observation language is not, re-teach the cropped-pot example.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Every green part is a leaf.

Stems and some flowers can also be green.

Check: Ask whether shape, attachment and position support the name.

No visible flower means it is not a flowering plant.

A flowering plant may not have open flowers at the time of observation.

Check: Compare no flower visible today with never flowers.

A pot with no visible roots has no roots.

An opaque pot can hide roots.

Check: Ask what the view allows us to observe.

Word	Meaning
root	A plant part usually below ground.
stem	The structure that connects and supports above-ground parts.
leaf	A usually flat plant part attached to a stem.
flower	A reproductive structure of a flowering plant.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

Science and Plants for Schools: plant biology animations

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.